

Mountless Aerial HMDs: Designing Non-Wearable HMD Experiences Based on Aerial Imaging Optics

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Figure 1: Overview of a mountless aerial HMD experience. Left: A conceptual illustration and prototype showing binocular VR content perceived as aerial images in front of the eyes without wearing a head-mounted device. Right: A prototype demonstration showing sufficient eyebox brightness and that the limited field of view of mountless aerial HMDs can be extended using a symmetrical mirror structure.

Abstract

Head-mounted displays (HMDs) have rapidly advanced in visual fidelity; however, improvements in resolution, field of view, and processing performance tend to inevitably lead to heavier and bulkier devices. Although various approaches have been proposed to balance visual quality and wearing comfort, this long-standing trade-off remains difficult to resolve as long as wearing a device is assumed. In this work, we explore non-wearable aerial HMD experiences based on aerial imaging optics, and propose a method for presenting binocular VR imagery as an aerial image in front of the user without requiring any head-mounted device. By relocating optical and computational components to the environment, the proposed systems reduce weight-induced discomfort while preserving immersive VR experiences. Furthermore, using a chair-integrated interaction design as one example, we investigate the design space of non-wearable HMDs and discuss how the proposed approach differs from conventional projection-based immersive displays and head-mounted displays.

CCS Concepts

- **Hardware** → Emerging optical and photonic technologies;
- **Human-centered computing** → Virtual reality.

ACM Reference Format:

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1 Introduction

To enable people to use virtual environments in a manner comparable to real-world interaction, natural and immersive experiences are essential. Head-mounted displays (HMDs), originating from Sutherland's seminal work in 1968 [Sutherland 1968], have served as the primary device for such experiences. Ideal visual requirements include a wide field of view, high resolution, high dynamic range, low latency, and depth presentation without vergence-accommodation conflict.

However, most existing HMDs rely on eyepiece- or relay-based optical systems [Kiyokawa 2007]. In such systems, expanding the field of view inevitably increases optical size and weight, resulting in a fundamental trade-off between image quality and wearing comfort. Even approaches such as beaming displays [Itoh et al. 2021], which separate the presentation and processing units, still require users to wear head-mounted components.



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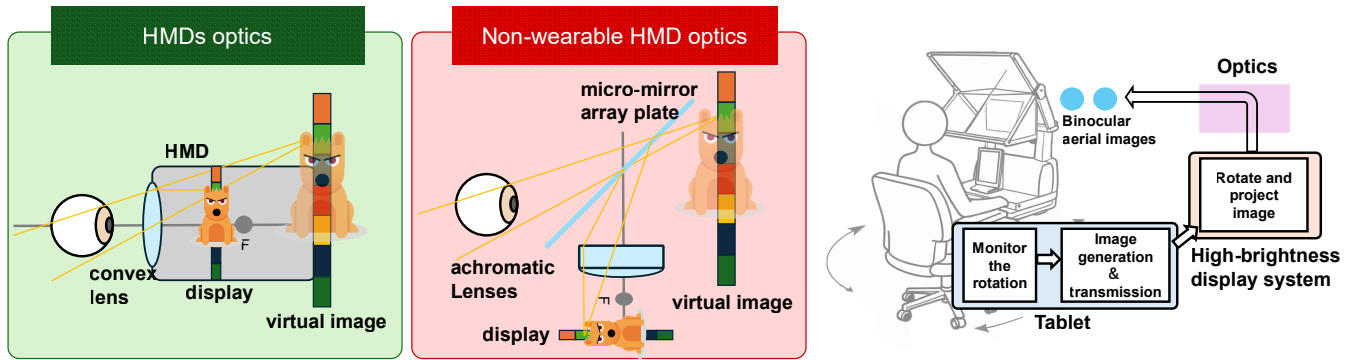


Figure 2: Implementation of the optical system and the chair-based system. Left: Comparison of the optical configurations between a conventional head-mounted display and the proposed mountless aerial HMD. Right: Block diagram of the newly developed chair-based system.

In parallel, the field of optics has explored alternative image formation methods that do not rely on head-mounted components. Retro-transmissive aerial imaging optical elements, which form images in front of a display surface, have been developed since around 2006 [Maekawa et al. 2006]. Based on this principle, the concept of false or virtual image projectors for head-mounted displays has also been proposed [Kijima and Watanabe 2009]. More recently, near-eye display systems based on aerial image projection [Otao et al. 2018] and aerial imaging by retro-reflection (AIRR) [Trovato et al. 2018] have demonstrated that combining retro-transmissive optical elements with convex lenses enables virtual images to be formed in front of the user without conventional eyepiece optics.

These prior studies provide important optical foundations for non-wearable and see-through near-eye displays. However, optical configurations alone are insufficient to achieve interactive experiences comparable to those realized by conventional HMDs. In particular, studies that employ practically implementable aerial imaging optics to create immersive interaction designs and that examine the characteristics of the design space of non-wearable HMDs through system prototyping remain limited.

To establish an alternative approach distinct from conventional projection-based dome screens and head-mounted displays, it is necessary to systematically explore the properties of such systems through practical implementations.

In this work, we focus on non-wearable HMDs based on aerial imaging optics and investigate interaction designs and system configurations that enable immersive VR experiences without assuming head-mounted devices. As one concrete example, we introduce a chair-integrated interaction design that couples the user’s bodily rotation with visual presentation, realizing a non-wearable VR experience. Through this implementation, we examine the characteristics of the design space of non-wearable HMDs and demonstrate the potential of this approach as an intermediate category between projection-based immersive displays and conventional head-mounted displays.

2 System and Experience Overview

The proposed mountless aerial HMD presents stereoscopic VR imagery as an aerial image in front of the user without requiring any device to be worn on the head. Figure 2 Left compares the optical configuration of a conventional HMD with that of the proposed system. By positioning a high-brightness display slightly beyond the focal plane of a convex (achromatic) lens, a real image is formed and folded back using a retro-transmissive optical element. Because this element attenuates luminance, a high-brightness display is employed, enabling a sufficiently bright virtual image to be perceived at a far distance in front of the user. This configuration achieves an optical ray path functionally equivalent to that of a conventional HMD while remaining entirely non-wearable.

To enable interactive VR experiences comparable to those of HMDs, we introduce a chair-integrated interaction design (Figure 2) Right. The optical module is rigidly attached to the seat and rotates with the user’s body. A vertically mounted tablet measures rotational motion via its built-in gyro sensor, allowing stereoscopic imagery to be generated and transmitted in real time. The field of view (FOV) is primarily determined by the size of the retro-transmissive optical element; however, as in prior aerial image displays, the viewing area can be extended by arranging a symmetrical mirror structure around the element [Kikuchi et al. 2023]. By placing planar mirrors above, below, and on both sides, the effective screen size can theoretically be increased by up to six times.

In the demonstration, visitors simply sit on a chair positioned in front of the system, without any headset fitting or calibration. Upon seating, users immediately perceive a stereoscopic virtual scene floating in midair at an appropriate viewing distance. The seamless transition between the physical environment and the virtual scene is a key characteristic of the proposed approach. The current system represents one implementation of transforming an HMD optical configuration into an aerial image display; because lens properties and size strongly influence experience quality, we envision future demonstrations with exchangeable components to further explore the design space of mountless aerial HMDs.

Table 1: Comparison with Existing Immersive Displays (○: well supported; △: conditionally supported; ×: structurally limited).

Design Aspect	Projection-Based	HMDs	Mountless Aerial HMDs
Non-wearable comfort	○	×	○
Freedom from room-size constraints	×	○	○
Wide image depth	×	○	○
Field of view	○	○	△
Large eyepiece / eye relief	○	×	○
System portability	×	○	△
Instant transition between physical and virtual spaces	○	×	○
Design position	Projection-dominant	HMD-dominant	Intermediate

3 Technical Contribution

One of the primary contributions of this work lies in reframing the HMD concept from the perspective of non-wearable, environment-anchored system design. Instead of optimizing wearable devices for reduced weight, we eliminate the requirement to wear an HMD.

Chair-integrated, non-wearable VR interaction. Beyond optical design, our system introduces a new interaction paradigm by integrating the VR viewing aperture into a chair-based structure. This configuration transforms the act of “looking into VR” into a seated, embodied experience, enabling interaction through natural rotational body movements. Because no head-mounted device is required, users transition into the VR environment immediately upon sitting down. This seamless and instantaneous shift between the physical environment and the virtual space constitutes a defining characteristic of the proposed interaction paradigm.

Aerial imaging with expanded field of view. The system employs a retro-transmissive aerial imaging optical configuration to form stereoscopic virtual images in free space. To improve experiential quality, we introduce a symmetrical mirror structure that expands the field of view. While prior work explored partial field-of-view expansion in aerial imaging [Kikuchi et al. 2023], balanced expansion in both vertical and horizontal directions for VR viewing has not been systematically addressed. Our design demonstrates how aerial imaging optics can support immersive VR viewing without head-mounted optics.

A third alternative for immersive displays. Traditionally, immersive visual experiences have been realized through two dominant approaches: projection-based immersive displays and wearable HMDs. This work suggests a third alternative—a non-wearable, aerial-image-based VR display that combines the environmental stability of projection systems with the personal viewpoint of HMDs. By doing so, it introduces a new category of immersive display that addresses long-standing limitations in comfort and deployability.

4 Design Space of Non-Wearable HMD Experiences

Table 1 summarizes the qualitative characteristics of existing immersive display paradigms and situates mountless aerial HMDs within this design space. Projection-based immersive displays provide non-wearable comfort and large eyepieces; however, image depth is constrained by room size and screen locations. In contrast,

head-mounted displays enable virtual images at flexible depths independent of the surrounding environment, but at the cost of wearing comfort.

While retaining the non-wearable comfort characteristic of projection-based systems, mountless aerial HMDs enable room-independent image depth presentation comparable to that of conventional HMDs. Furthermore, an important feature of this approach is that transitions between immersive and non-immersive states are not controlled by wearing or removing a device, but instead by physically entering or leaving the aerial image volume (e.g., a floating luminous sphere). This enables smooth and continuous transitions between physical and virtual spaces.

5 Conclusion

This work explores an alternative approach to immersive VR that removes the assumption of wearing a head-mounted device. By presenting stereoscopic imagery as an aerial image in front of the user, the proposed system preserves immersive visual qualities while eliminating weight-induced discomfort. This approach introduces a new design space for non-wearable HMD experiences.

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